The user manual of GS-Impute v1.0

System Requirements:

PyTorch installation (mandatory)

GPU-enabled PyTorch version (recommended)

Here are the install commands of PyTorch. Scroll down for the step-by-step instructions.

1 Step-by-step instructions of installing PyTorch

1.1 Install Miniconda

Miniconda is the recommended approach for installing PyTorch with GPU support. It creates a separate environment to avoid changing any installed software in your system. This is also the easiest way to install the required software especially for the GPU setup. Download the Miniconda Windows Installer (*https://repo.anaconda.com/miniconda/Miniconda3-latest-Windows-x86_64.exe*) or the Miniconda Linux Installer (*https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86.sh*). Follow the on-screen instructions and choose the default options.

1.2 Create a Conda environment

Open a terminal (by clicking on the “Anaconda Promp” button through the Start menu in Windows). Create a Conda virtual environment named pytorch (other names are also acceptable) with the following commands in a terminal window (taking python 3.9 as an example).

conda create --name pytorch python=3.9

You can deactivate and activate it with the following commands.

conda deactivate

conda activate pytorch

Make sure it is activated for the rest of the installation.

1.3 Install PyTorch

Install PyTorch (with GPU support) using the following command (taking NVIDIA RTX 4090 as an example).

pip3 install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118

If you are not using a GPU (not recommended), run the following command.

pip3 install torch torchvision torchaudio

2 Install GS-Impute

You should download the GSImpute v1.0_Windows package. If under the Windows operating system, open a terminal window and navigate to the download directory (taking “D:\GSImpute v1.0_Windows” as an example). Then activate your Conda environment that has already installed PyTorch, and install GS-Impute with the following commands.

D:

cd GSImpute v1.0_Windows

conda activate pytorch
python install.py

3 Usage

3.1 For the Windows operating system

In a Windows operating system, a typical command for running GS-Impute is as follows:

gsi

Once you execute the command above in a terminal window, general imputation screen will be opened by default. The software interface and input/output format examples are shown in Fig. 1.

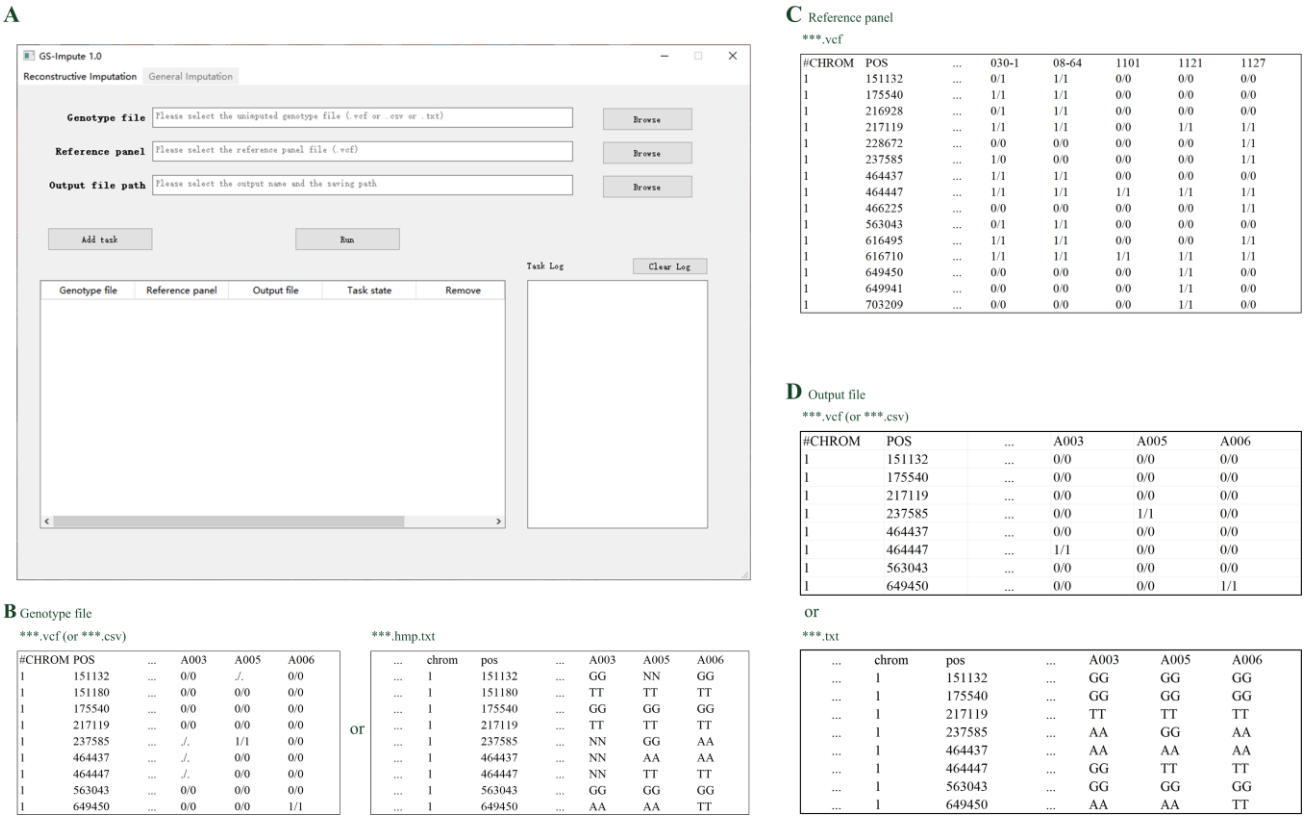

Fig. 1. Software interface in the Windows system and input/output format examples of GS-Impute 1.0 for general imputation. (A) User interface of GS-Impute 1.0 for general imputation in the Windows system. **(B)** Example of the genotype file format used as input for general imputation, where "..." represents omitted nonessential information. The file may contain markers from multiple chromosomes. For a genotype files in CSV or TXT format, the column header immediately preceding data must be ref, alt, or QCcode. **(C)** Example of the reference panel format used as input for general imputation. The panel must be a VCF file corresponding to a single chromosome, and only markers on that chromosome can be imputed. Missing markers are not permitted in the reference panel. **(D)** Example of the output file format for general imputation.

You can click the "*Browse*" button on the right side to select the genotype file, the reference panel and the output file path.

Once you have selected all the files, you can click the "*Add task*" button to add general imputation tasks to the task list (Fig. 2).

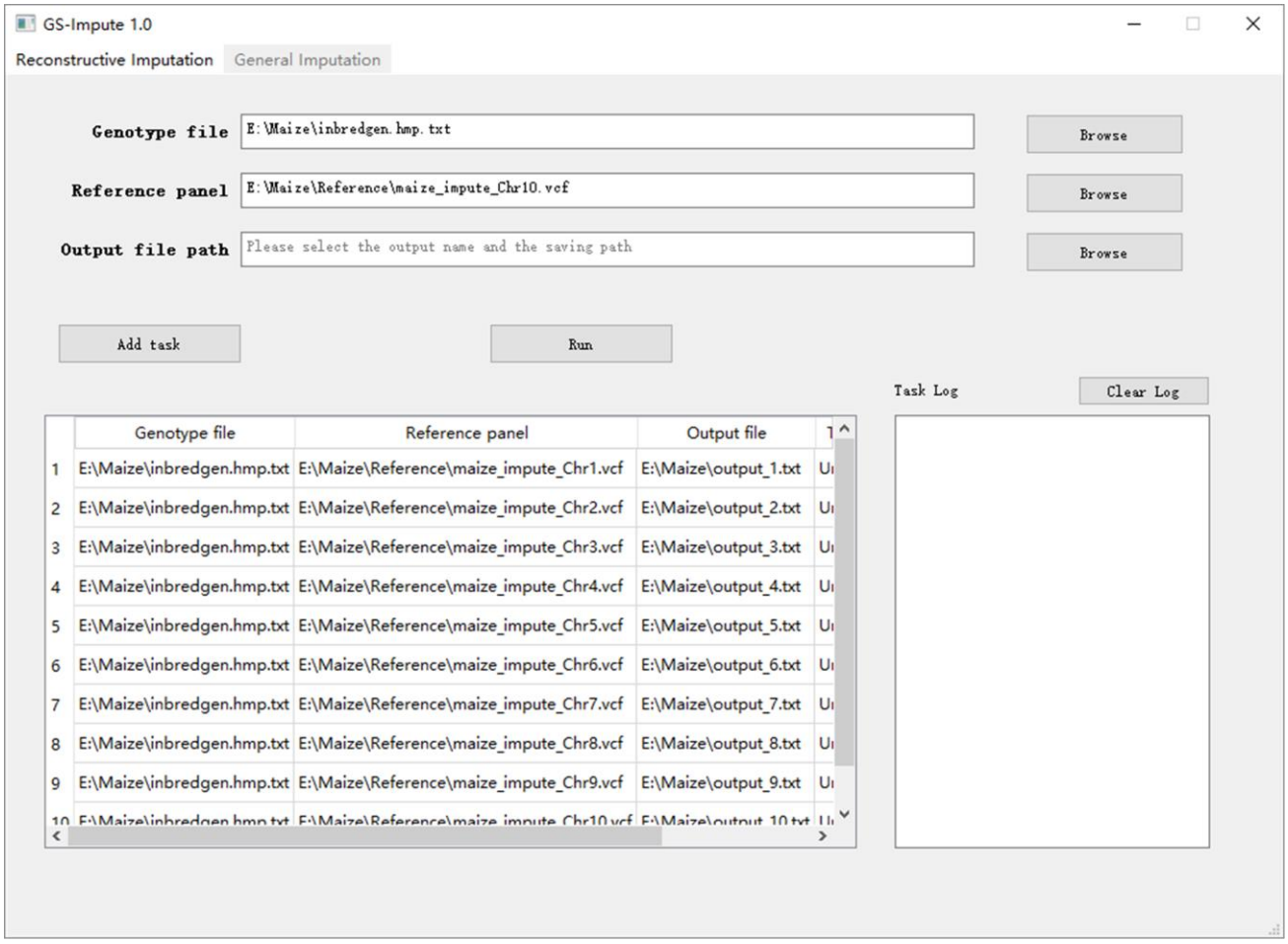

Fig. 2. Software interface of GS-Impute 1.0 with general imputation tasks added

If all tasks have been added, you can click the "*Run*" button to execute the tasks. GS-Impute will execute tasks in the order of the task list.

During task execution, the task log on the right shows real-time progress and detailed information of the tasks.

You can also click the "*Reconstructive Imputation*" button to perform your reconstructive imputation (Fig. 3).

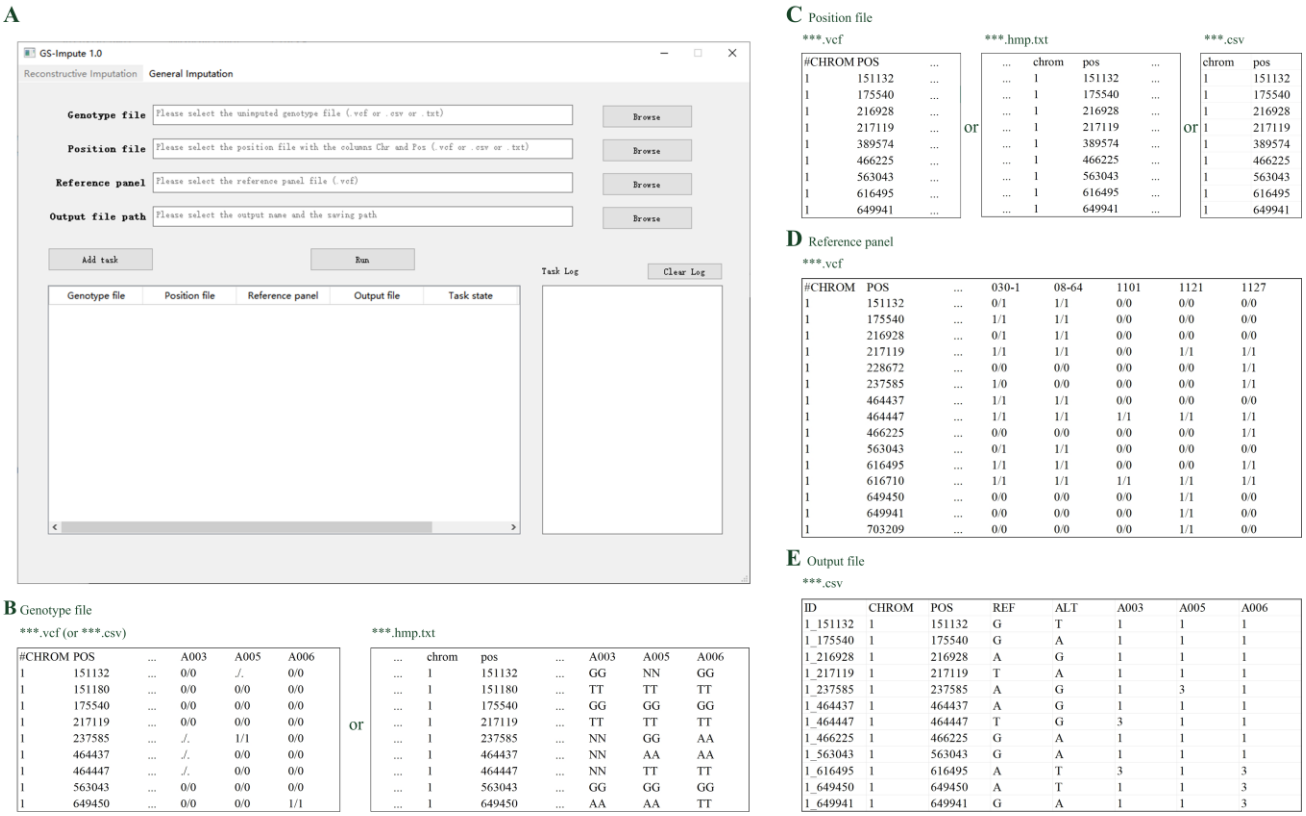

Fig. 3. Software interface in the Windows system and input/output format examples of GS-Impute 1.0 for reconstructive imputation. (A) User interface of GS-Impute 1.0 for reconstructive imputation in the Windows system. **(B)** Example of the genotype file format used as input for reconstructive imputation, where "..." represents omitted nonessential information. The file may contain markers from multiple chromosomes. For a genotype files in CSV or TXT format, the column header immediately preceding data must be ref, alt, or QCcode. **(C)** Example of the position file format used as input for reconstructive imputation. The file may contain position information from multiple chromosomes. A genotype file containing chromosome and position information can also be used as the position file. **(D)** Example of the reference panel format used as input for reconstructive imputation. The panel must be a VCF file corresponding to a single chromosome, and only markers on that chromosome can be imputed. Missing markers are not permitted in the reference panel. **(E)** Example of the output file format for reconstructive imputation. Genotype values of 1 and 3 represent homozygous variants with two reference alleles and two alternate alleles, respectively. The genotype value of 2 represents a heterozygous variant containing one reference allele and one alternate allele.

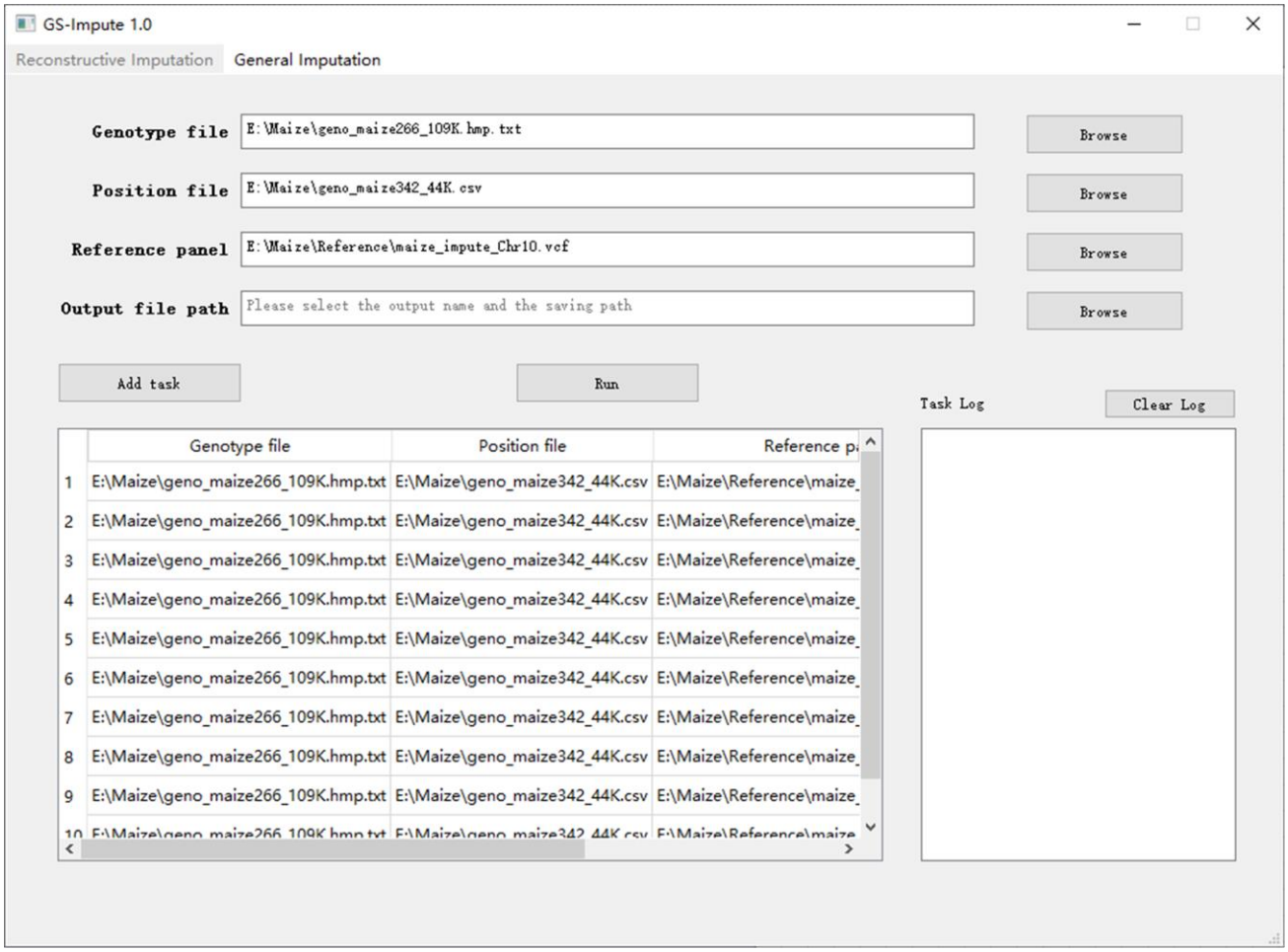

Fig. 4. Software interface of GS-Impute 1.0 with reconstructive imputation tasks added

3.2 For the Linux operating system

A typical GS-Impute command for the general imputation is as follows:

gsi -i -geno unimputed_geno_file.vcf -panel panel_file.vcf

Or

gsi -i -geno unimputed_geno_file.csv -panel panel_file.vcf

Or

gsi -i -geno unimputed_geno_file.hmp.txt -panel panel_file.vcf

A typical GS-Impute command for the reconstructive imputation is as follows:

gsi -r -geno unimputed_geno_file.vcf -pos position_file.vcf -panel panel_file.vcf

Or

gsi -r -geno unimputed_geno_file.csv -pos position_file.vcf -panel panel_file.vcf

Or

gsi -r -geno unimputed_geno_file.hmp.txt -pos position_file.vcf -panel panel_file.vcf

Or

gsi -r -geno unimputed_geno_file.vcf -pos position_file.hmp.txt -panel panel_file.vcf

Or

gsi -r -geno unimputed_geno_file.csv -pos position_file.hmp.txt -panel panel_file.vcf

Or

gsi -r -geno unimputed_geno_file.hmp.txt -pos position_file.hmp.txt -panel panel_file.vcf

Or

gsi -r -geno unimputed_geno_file.vcf -pos position_file.csv -panel panel_file.vcf

Or

gsi -r -geno unimputed_geno_file.csv -pos position_file.csv -panel panel_file.vcf

Or

gsi -r -geno unimputed_geno_file.hmp.txt -pos position_file.csv -panel panel_file.vcf

The file names of *unimputed_geno_file*, *position_file* and *panel_file* (listed above) can be freely defined, but their formats must strictly conform to the requirements specified in Fig. 1 and Fig. 3.

GS-Impute allows batch execution of imputation tasks via shell scripts in Linux. Please store the unimputed genotype files and the reference panels in the same directory, and then create a shell script file, such as *gsi.sh*. Type the imputation commands following the example below, and then save it.

gsi -i -geno unimputed_geno_file1.vcf -panel panel_file1.vcf -out outputfile1

gsi -i -geno unimputed_geno_file2.vcf -panel panel_file2.vcf -out outputfile2

Open the terminal in the current directory. Activate the virtual environment “pytorch” and use the bash command to execute the shell script.

bash gsi.sh

The program will perform the imputation tasks in the order of the shell script, and the imputed files will be output in the current directory.